



EXPLOITS UNIVERSITY

BSc INFORMATION COMMUNICATION AND TECHNOLOGY

BICT 3202 · OBJECT-ORIENTED ANALYSIS AND DESIGN

WEEK 2

Assignment: Principles of Object Modelling

PROGRAMME	BSc Information Communication and Technology
COURSE CODE / YEAR	BICT 3202 · Year 3
LECTURER	Francis Fweta — ICT Lecturer
DELIVERY	2h Lecture · 1h Tutorial · 1h Practical · 10h Independent Learning

Prepared by Francis Fweta · Exploits University · Week 2 of 16



Contents

ASSIGNMENT INSTRUCTIONS	3
Section A — Short-Answer Questions (20 marks)	3
Section B — Application Questions (20 marks)	4
Section C — Case Study (10 marks)	4
Marking Scheme	5
Submission & Academic Integrity	5

ASSIGNMENT INSTRUCTIONS

- Answer ALL questions in Sections A, B and C.
- Read the Week 2 module and your lecture notes before attempting each question.
- Where a question asks you to 'identify', write clear, specific items and justify each choice.
- Cite ISO/IEC/IEEE 29148:2018 and OMG UML 2.5.1 where relevant.
- Submit by the date given by your lecturer. Total: 50 marks.

Section A — Short-Answer Questions (20 marks)

Question A1

Define object modelling. [2 marks]

Question A2

Define an object model. [2 marks]

Question A3

Why are object models useful in systems analysis? [2 marks]

Question A4

Explain the relationship between an object, a class and an object model. [2 marks]

Question A5

Define abstraction and give one example from object modelling. [2 marks]

Question A6

Define encapsulation and explain why it matters for a bank account. [2 marks]

Question A7

What is object identity? How does it differ from an attribute? [2 marks]

Question A8

Distinguish between object state and object behaviour. [2 marks]

Question A9

What is object identification? Why should not every noun become a class? [2 marks]

Question A10

Name the standard language used to express object models, and state the current published specification version. [2 marks]

Section B — Application Questions (20 marks)

Question B1 (12 marks)

'A customer uses a mobile phone to purchase a product from an online shop. The customer adds the product to a shopping cart and pays using mobile money. The system generates an invoice.'

Identify:

- (a) six candidate objects; [3 marks]
- (b) two attributes for three of the objects; [3 marks]
- (c) two behaviours for two of the objects; [2 marks]
- (d) three relationships; [2 marks]
- (e) two external systems, and state whether 'Mobile Phone', 'Mobile Money' and 'Invoice' should be objects — with reasons. [2 marks]

Question B2 (8 marks)

A hospital wants an electronic appointment management system. Patients request appointments with doctors; doctors have specialties; receptionists manage appointments.

- (a) five candidate objects; [2 marks]
- (b) attributes for Patient and Appointment; [2 marks]
- (c) three behaviours; [2 marks]
- (d) two relationships, and one piece of information to be excluded through abstraction. [2 marks]

Section C — Case Study (10 marks)

'A university wants to develop an online course registration system. Students select courses and submit a registration; lecturers approve registrations; the system enforces prerequisites and records payments.'

As an object-oriented analyst:

- (a) Identify ten candidate objects. [2 marks]
- (b) Give five attributes for appropriate objects and five behaviours. [3 marks]
- (c) Give five relationships between objects. [2 marks]
- (d) Explain how abstraction, encapsulation and modularity would help you keep this model simple and maintainable. [3 marks]



Marking Scheme

Section	Content	Marks
A	Short-answer questions (10 × 2)	20
B	Application questions (12 + 8)	20
C	Case study	10
	TOTAL	50

Submission & Academic Integrity

- Submit a typed document (or neatly handwritten, as directed by your lecturer).
- This assignment is individual work. Plagiarism or copying will attract a penalty under university regulations.
- Cite any sources you consult using the format used in the Week 2 module references.